VISVESVARAYA TECHNOLOGICAL UNIVERSITY

"JnanaSangama", Belgaum -590014, Karnataka.

LAB REPORT

On

ANALYSIS AND DESIGN OF ALGORITHMS (23CS4PCADA)

Submitted by

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in partial fulfillment for the award of the degree of
BACHELOR OF ENGINEERING
in
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B.M.S. COLLEGE OF ENGINEERING
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CERTIFICATE

B. M. S. College of Engineering,
Bull Temple Road, Bangalore 560019
(Affiliated To Visvesvaraya Technological University, Belgaum) Department of Computer Science and Engineering

This is to certify that the Lab work entitled **"ANALYSIS AND DESIGN OF ALGORITHMS"** carried out by **KRITARTH TELI (1BF24CS150)**, who is bonafide student of **B. M. S. College of Engineering**. It is in partial fulfillment for the award of **Bachelor of Engineering in Computer Science and Engineering** of the Visvesvaraya Technological University, Belgaum during the year 2025-26. The Lab report has been approved as it satisfies the academic requirements in respect of Analysis and Design of Algorithms Lab - **(23CS4PCADA)** work prescribed for the said degree.

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Lab program 1:
Write program to obtain the Topological ordering of vertices in a given digraph.

Code:

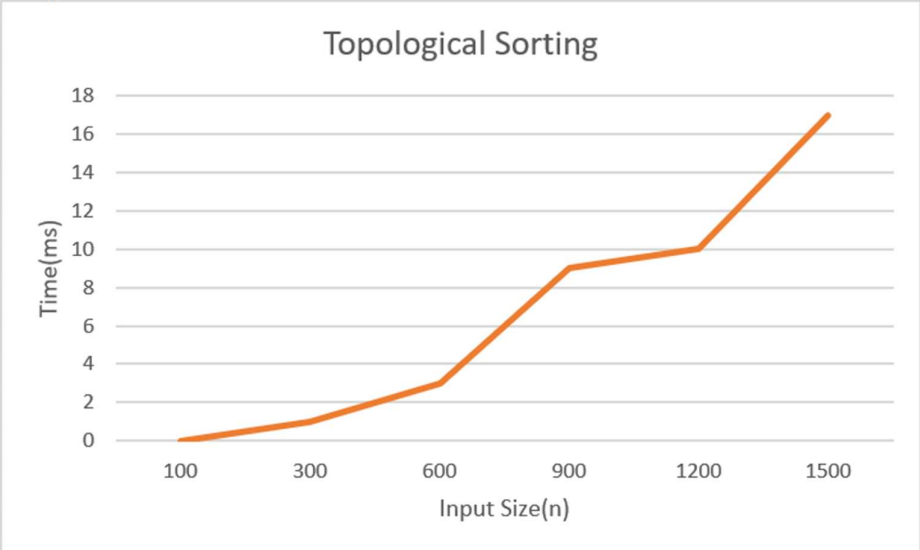
```
#include <stdio.h>
#define MAX 100

int graph[MAX][MAX];
int visited[MAX];
int stack[MAX];
int top = -1;
int n;

void DFS(int v) {
   visited[v] = 1;

   for(int i = 0; i < n; i++)
      if(graph[v][i] == 1 && !visited[i])
         DFS(i);

   stack[++top] = v;
}

void topologicalSort() {
   for(int i = 0; i < n; i++)
      if(!visited[i])
         DFS(i);

   printf("\nTopological Ordering:\n");

   while(top != -1)
      printf("%d ", stack[top--]);
}

int main() {
   int edges, u, v;
   printf("Enter number of vertices: ");
   scanf("%d", &n);
   printf("Enter number of edges: ");
   scanf("%d", &edges);
```

```
    for(int i = 0; i < n; i++)
        for(int j = 0; j < n; j++)
            graph[i][j] = 0;

    printf("Enter edges (u v):\n");

    for(int i = 0; i < edges; i++) {
        scanf("%d %d", &u, &v);
        graph[u][v] = 1;
    }

    topologicalSort();
    return 0;
}
```

OUTPUT:

```
Enter number of vertices: 4
Enter number of edges: 3
Enter edges (u v):
0 1
1 2
2 3

Topological Ordering:
0 1 2 3
Process returned 0 (0x0)   execution time : 16.799 s
Press any key to continue.
```

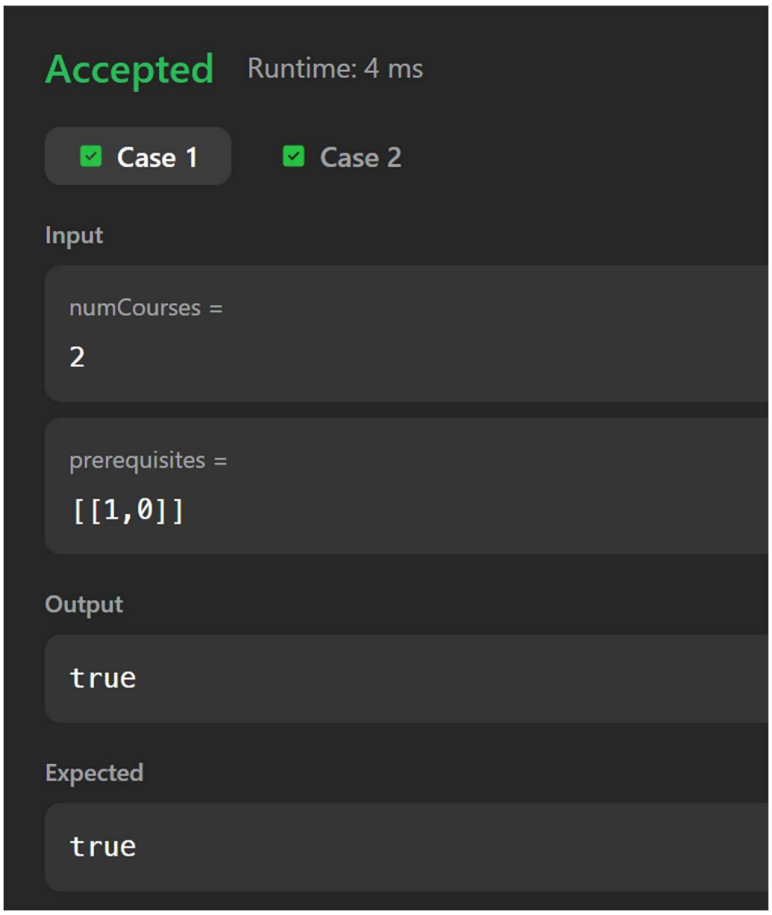
GRAPH:



LeetCode 207 - Course Schedule

```
bool canFinish(int numCourses, int** prerequisites, int prerequisitesSize,
                    int* prerequisitesColSize) {
    int graph[2000][2000] = {0};
    int indegree[2000] = {0};
    int size[2000] = {0};

    for(int i = 0; i < prerequisitesSize; i++) {
        int a = prerequisites[i][0];
        int b = prerequisites[i][1];
        graph[b][size[b]++] = a;
        indegree[a]++;
    }
    int queue[2000];
    int front = 0, rear = 0;

    for(int i = 0; i < numCourses; i++)
        if(indegree[i] == 0)
            queue[rear++] = i;
    int count = 0;

    while(front < rear) {
        int node = queue[front++];
        count++;

        for(int i = 0; i < size[node]; i++) {
            int next = graph[node][i];
            indegree[next]--;
            if(indegree[next] == 0)
                queue[rear++] = next;
        }
    }

    return count == numCourses;
}
```

OUTPUT:


Accepted
Runtime: 4 ms
Case 1
Case 2
Input
numCourses =
2
prerequisites =
[[1,0]]
Output
true
Expected
true

Lab program 2:
Implement Johnson Trotter algorithm to generate permutations.

Code:

```
#include <stdio.h>
#define LEFT -1
#define RIGHT 1

void printPermutation(int a[], int n) {
   for(int i = 0; i < n; i++)
      printf("%d ", a[i]);
   printf("\n");
}

int getMobile(int a[], int dir[], int n) {
   int mobile = 0, pos = -1;

   for(int i = 0; i < n; i++) {
      if(dir[i] == LEFT && i != 0 && a[i] > a[i - 1]) {
         if(a[i] > mobile) {
            mobile = a[i];
            pos = i;
         }
      }

      if(dir[i] == RIGHT && i != n - 1 && a[i] > a[i + 1]) {
         if(a[i] > mobile) {
            mobile = a[i];
            pos = i;
         }
      }
   }

   return pos;
}

void johnsonTrotter(int n) {
   int a[n], dir[n];

   for(int i = 0; i < n; i++) {
      a[i] = i + 1;
      dir[i] = LEFT;
   }
```

```
   printPermutation(a, n);

   while(1) {
      int pos = getMobile(a, dir, n);
      if(pos == -1)
         break;
      int swapPos;
      if(dir[pos] == LEFT)
         swapPos = pos - 1;
      else
         swapPos = pos + 1;
      int temp = a[pos];
      a[pos] = a[swapPos];
      a[swapPos] = temp;
      temp = dir[pos];
      dir[pos] = dir[swapPos];
      dir[swapPos] = temp;
      pos = swapPos;
      for(int i = 0; i < n; i++)
         if(a[i] > a[pos])
            dir[i] = -dir[i];
      printPermutation(a, n);
   }
}

int main() {
   int n;
   printf("Enter n: ");
   scanf("%d", &n);
   johnsonTrotter(n);

   return 0;
}
```

OUTPUT:

```
Enter n: 3
1 2 3
1 3 2
3 1 2
3 2 1
2 3 1
2 1 3

Process returned 0 (0x0)   execution time : 33.420 s
Press any key to continue.
```

Lab program 3:
Sort a given set of N integer elements using Merge Sort technique and compute its time taken. Run the program for different values of N and record the time taken to sort.

Code:

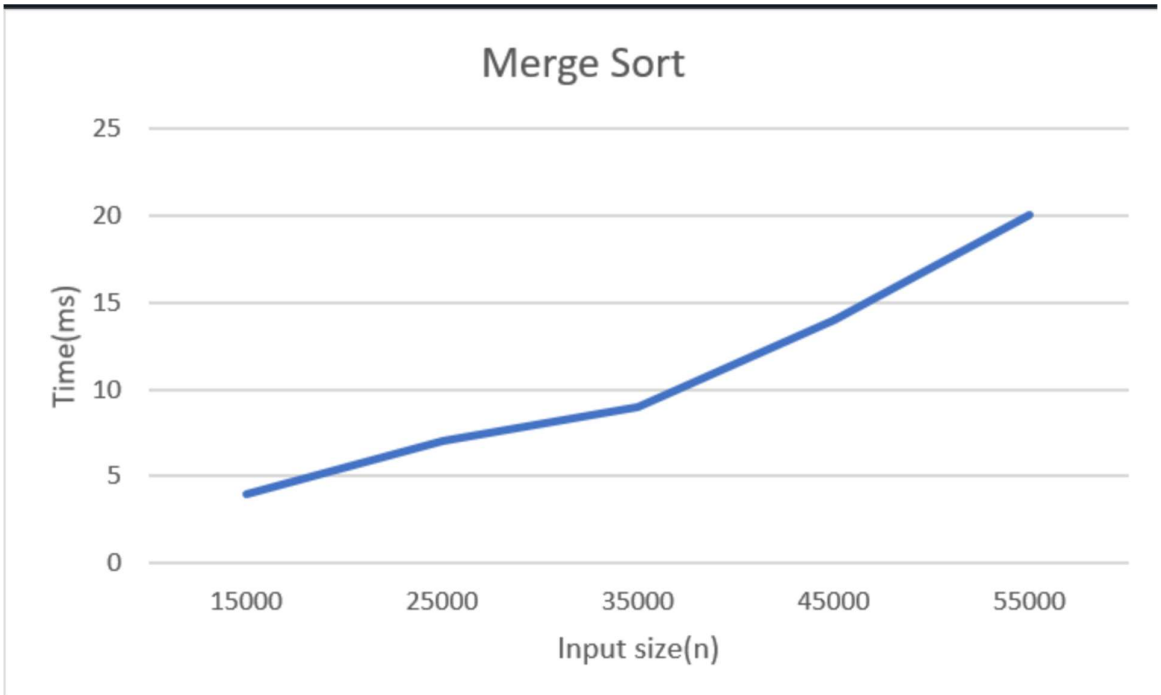
```
#include <stdio.h>
#include <stdlib.h>
#include <time.h>

void merge(int a[], int low, int mid, int high) {
   int i = low, j = mid + 1, k = 0;
   int temp[100000];

   while(i <= mid && j <= high) {
      if(a[i] < a[j])
         temp[k++] = a[i++];
      else
         temp[k++] = a[j++];
   }

   while(i <= mid)
      temp[k++] = a[i++];

   while(j <= high)
      temp[k++] = a[j++];

   for(i = low, k = 0; i <= high; i++, k++)
      a[i] = temp[k];
}

void mergeSort(int a[], int low, int high) {
   if(low < high) {
      int mid = (low + high) / 2;
      mergeSort(a, low, mid);
      mergeSort(a, mid + 1, high);
      merge(a, low, mid, high);
   }
}
```

```
int main() {
    int n;
    int a[100000];
    printf("Enter number of elements: ");
    scanf("%d", &n);
    for(int i = 0; i < n; i++)
        a[i] = rand() % 1000;
    clock_t start, end;
    start = clock();
    mergeSort(a, 0, n - 1);
    end = clock();
    double time_taken = (double)(end - start) / CLOCKS_PER_SEC;
    printf("\nSorted Elements:\n");

    for(int i = 0; i < n; i++)
        printf("%d ", a[i]);

    printf("\n\nTime Taken = %f seconds\n", time_taken);

    return 0;
}
```

OUTPUT:

```
Enter number of elements: 10

Sorted Elements:
41 169 334 358 464 467 478 500 724 962

Time Taken = 0.000000 seconds

Process returned 0 (0x0)   execution time : 32.283 s
Press any key to continue.
```

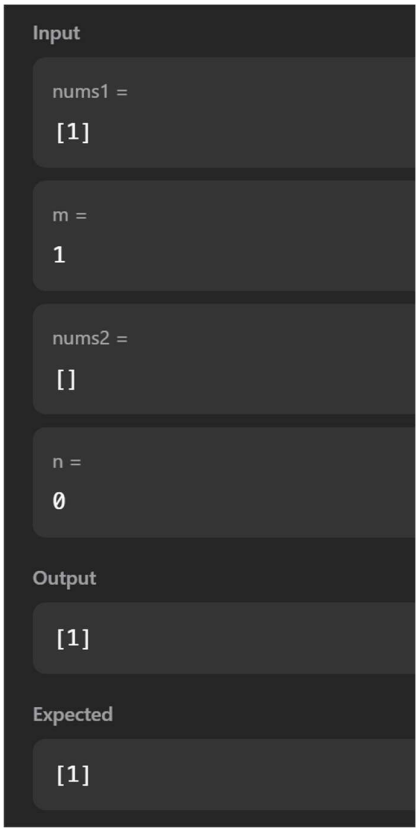
GRAPH:



LeetCode 88 - Merge Sorted Array

```
void merge(int* nums1, int m, int* nums2, int n) {
    int i = m - 1;
    int j = n - 1;
    int k = m + n - 1;

    while(i >= 0 && j >= 0) {
        if(nums1[i] > nums2[j])
            nums1[k--] = nums1[i--];
        else
            nums1[k--] = nums2[j--];
    }

    while(j >= 0)
        nums1[k--] = nums2[j--];
}
```

OUTPUT:



Lab program 4:
Sort a given set of N integer elements using Quick Sort technique and compute its time taken.

Code:

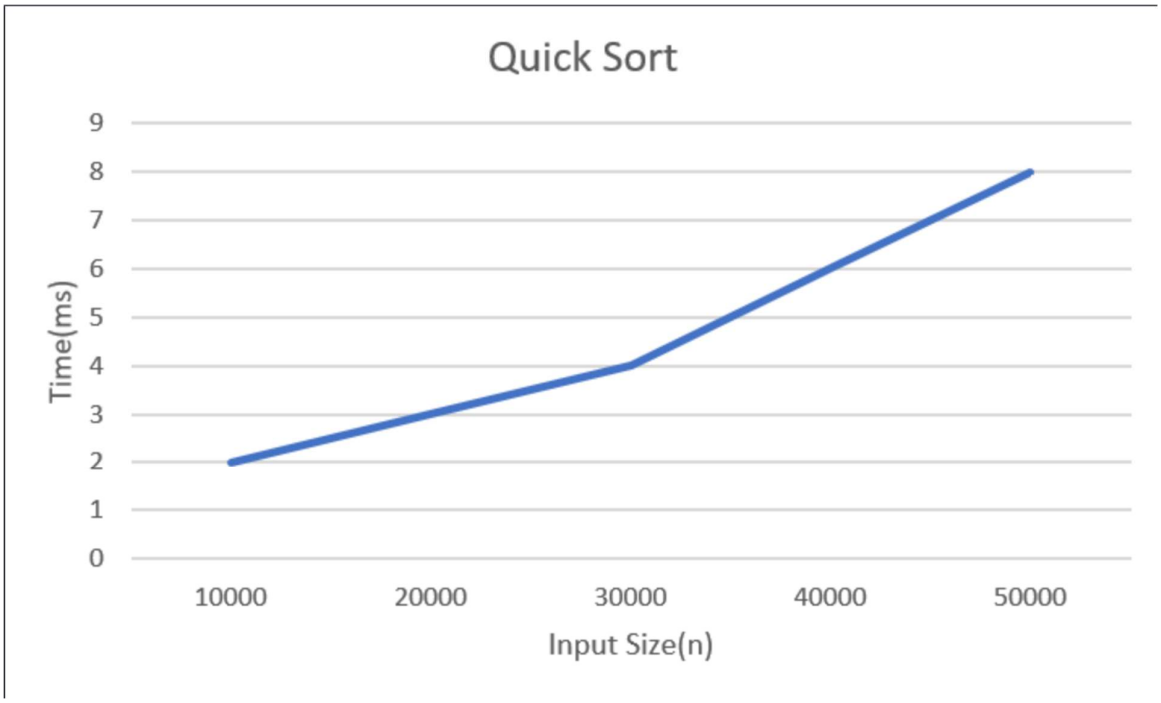
```
#include <stdio.h>
#include <stdlib.h>
#include <time.h>

void swap(int *a, int *b) {
   int temp = *a;
   *a = *b;
   *b = temp;
}

int partition(int a[], int low, int high) {
   int pivot = a[low];
   int i = low + 1;
   int j = high;

   while(i <= j) {
      while(i <= high && a[i] <= pivot)
         i++;
      while(a[j] > pivot)
         j--;
      if(i < j)
         swap(&a[i], &a[j]);
   }

   swap(&a[low], &a[j]);
   return j;
}

void quickSort(int a[], int low, int high) {
   if(low < high) {
      int p = partition(a, low, high);
      quickSort(a, low, p - 1);
      quickSort(a, p + 1, high);
   }
}
```

```
int main() {
    int n;
    int a[100000];
    printf("Enter number of elements: ");
    scanf("%d", &n);

    for(int i = 0; i < n; i++)
        a[i] = rand() % 1000;

    clock_t start, end;
    start = clock();
    quickSort(a, 0, n - 1);
    end = clock();
    double time_taken = (double)(end - start) / CLOCKS_PER_SEC;
    printf("\nSorted Elements:\n");

    for(int i = 0; i < n; i++)
        printf("%d ", a[i]);

    printf("\n\nTime Taken = %f seconds\n", time_taken);

    return 0;
}
```

OUTPUT:

```
Enter number of elements: 10

Sorted Elements:
41 169 334 358 464 467 478 500 724 962

Time Taken = 0.000000 seconds

Process returned 0 (0x0)   execution time : 2.523 s
Press any key to continue.
```

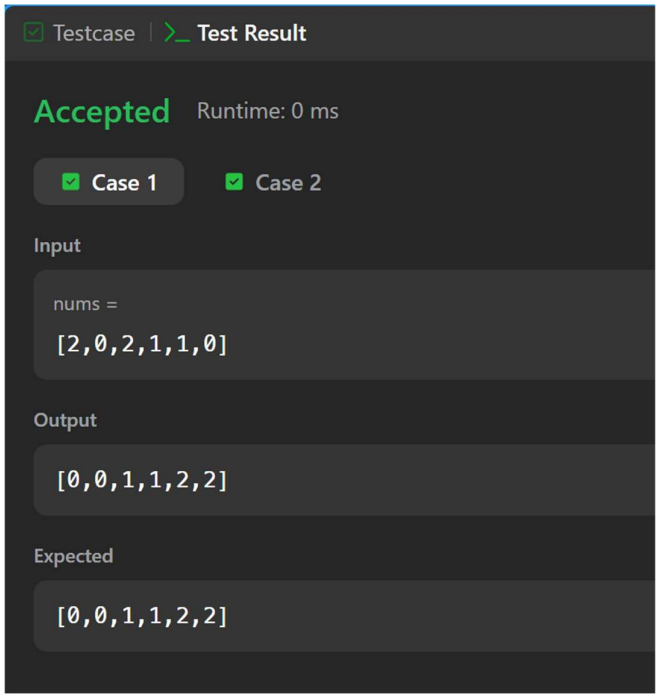
GRAPH:



LeetCode 75 - Sort Colors

```
void sortColors(int* nums, int numsSize) {
    int low = 0, mid = 0, high = numsSize - 1;

    while(mid <= high) {
        if(nums[mid] == 0) {
            int temp = nums[low];
            nums[low] = nums[mid];
            nums[mid] = temp;
            low++;
            mid++;
        }
        else if(nums[mid] == 1)
            mid++;
        else {
            int temp = nums[mid];
            nums[mid] = nums[high];
            nums[high] = temp;
            high--;
        }
    }
}
```

OUTPUT:



Lab program 5:
Sort a given set of N integer elements using Heap Sort technique and compute its time taken.

Code:

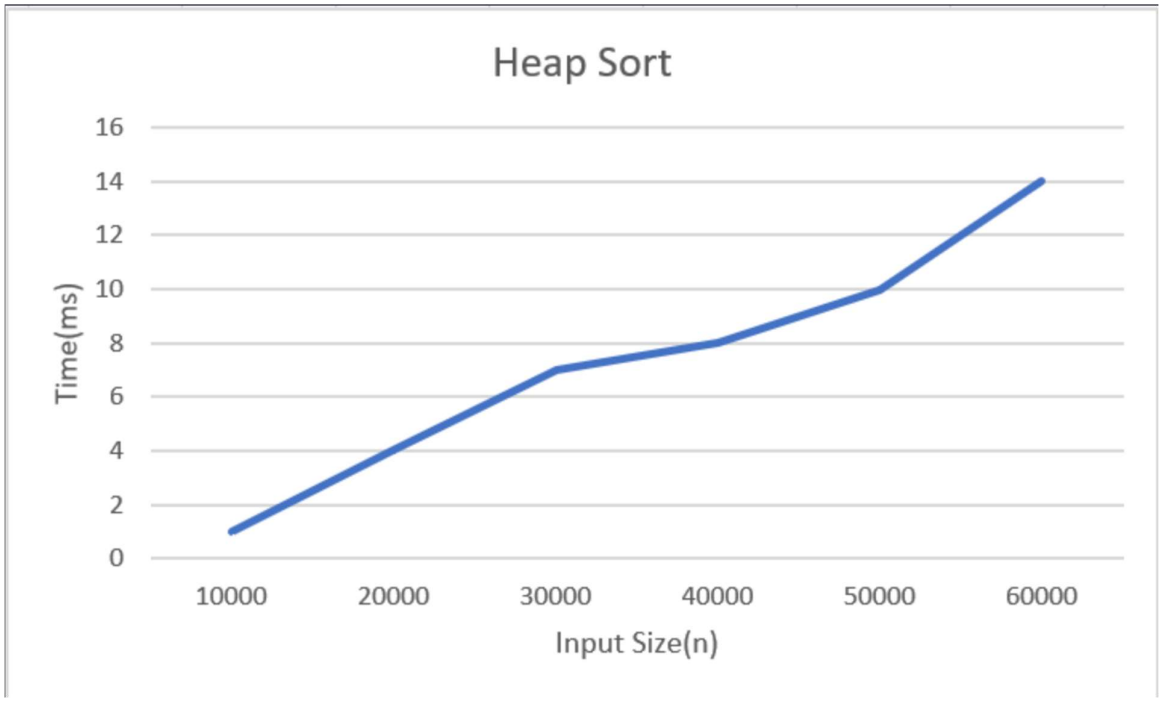
```
#include <stdio.h>
#include <stdlib.h>
#include <time.h>

void swap(int *a, int *b) {
   int temp = *a;
   *a = *b;
   *b = temp;
}

void heapify(int a[], int n, int i) {
   int largest = i;
   int left = 2 * i + 1;
   int right = 2 * i + 2;

   if(left < n && a[left] > a[largest])
      largest = left;

   if(right < n && a[right] > a[largest])
      largest = right;

   if(largest != i) {
      swap(&a[i], &a[largest]);
      heapify(a, n, largest);
   }
}

void heapSort(int a[], int n) {
   for(int i = n / 2 - 1; i >= 0; i--)
      heapify(a, n, i);

   for(int i = n - 1; i > 0; i--) {
      swap(&a[0], &a[i]);
      heapify(a, i, 0);
   }
}
```

```
int main() {
    int n;
    int a[100000];
    printf("Enter number of elements: ");
    scanf("%d", &n);

    for(int i = 0; i < n; i++)
        a[i] = rand() % 1000;

    clock_t start, end;
    start = clock();
    heapSort(a, n);
    end = clock();
    double time_taken = (double)(end - start) / CLOCKS_PER_SEC;
    printf("\nSorted Elements:\n");

    for(int i = 0; i < n; i++)
        printf("%d ", a[i]);

    printf("\n\nTime Taken = %f seconds\n", time_taken);

    return 0;
}
```

OUTPUT:

```
Enter number of elements: 10

Sorted Elements:
41 169 334 358 464 467 478 500 724 962

Time Taken = 0.000000 seconds

Process returned 0 (0x0)   execution time : 8.672 s
Press any key to continue.
```

GRAPH:



Lab program 6:
Implement 0/1 Knapsack problem using dynamic programming.

Code:

```
#include <stdio.h>

int max(int a, int b) {
   if(a > b)
      return a;
   return b;
}

int main() {
   int n, W;
   printf("Enter number of items: ");
   scanf("%d", &n);
   int wt[n], val[n];
   printf("Enter weights:\n");

   for(int i = 0; i < n; i++)
      scanf("%d", &wt[i]);

   printf("Enter profits:\n");

   for(int i = 0; i < n; i++)
      scanf("%d", &val[i]);

   printf("Enter capacity of knapsack: ");
   scanf("%d", &W);
   int dp[n + 1][W + 1];

   for(int i = 0; i <= n; i++) {
      for(int w = 0; w <= W; w++) {
         if(i == 0 || w == 0)
            dp[i][w] = 0;
         else if(wt[i - 1] <= w)
            dp[i][w] = max(val[i - 1] + dp[i - 1][w - wt[i - 1]], dp[i - 1][w]);
         else
            dp[i][w] = dp[i - 1][w];
      }
   }

   printf("\nMaximum Profit = %d\n", dp[n][W]);
   return 0;
}
```

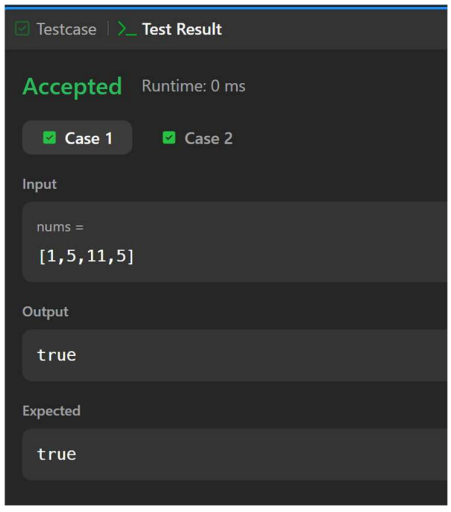
OUTPUT:

```
Enter number of items: 3
Enter weights:
10 20 30
Enter profits:
60 100 120
Enter capacity of knapsack: 50

Maximum Profit = 220

Process returned 0 (0x0)   execution time : 23.235 s
Press any key to continue.
```

LeetCode 416 - Partition Equal Subset Sum

```
bool canPartition(int* nums, int numsSize) {
    int sum = 0;

    for(int i = 0; i < numsSize; i++)
        sum += nums[i];

    if(sum % 2 != 0)
        return false;

    int target = sum / 2;
    bool dp[target + 1];

    for(int i = 0; i <= target; i++)
        dp[i] = false;

    dp[0] = true;

    for(int i = 0; i < numsSize; i++) {
        for(int j = target; j >= nums[i]; j--) {
            if(dp[j - nums[i]])
                dp[j] = true;
        }
    }

    return dp[target];
}
```

OUTPUT:



Lab program 7:
Implement All Pair Shortest paths problem using Floyd's algorithm.

CODE:

```
#include <stdio.h>
#define INF 99999

void floydWarshall(int graph[][10], int V) {
   int dist[10][10];

   for(int i = 0; i < V; i++) {
      for(int j = 0; j < V; j++)
         dist[i][j] = graph[i][j];
   }

   for(int k = 0; k < V; k++) {
      for(int i = 0; i < V; i++) {
         for(int j = 0; j < V; j++) {
            if(dist[i][k] != INF && dist[k][j] != INF && dist[i][k] + dist[k][j] < dist[i][j])
               dist[i][j] = dist[i][k] + dist[k][j];

         }
      }
   }
   printf("Shortest Distance Matrix:\n");

   for(int i = 0; i < V; i++) {
      for(int j = 0; j < V; j++) {
         if(dist[i][j] == INF)
            printf("INF ");
         else
            printf("%3d ", dist[i][j]);
      }
      printf("\n");
   }
}

int main() {
   int V;
   printf("Enter number of vertices: ");
   scanf("%d", &V);
   int graph[10][10];
   printf("Enter adjacency matrix:\n");
   printf("(Enter %d for INF)\n", INF);
```

```
    for(int i = 0; i < V; i++) {
        for(int j = 0; j < V; j++)
            scanf("%d", &graph[i][j]);
    }
    floydWarshall(graph, V);

    return 0;
}
```

OUTPUT:

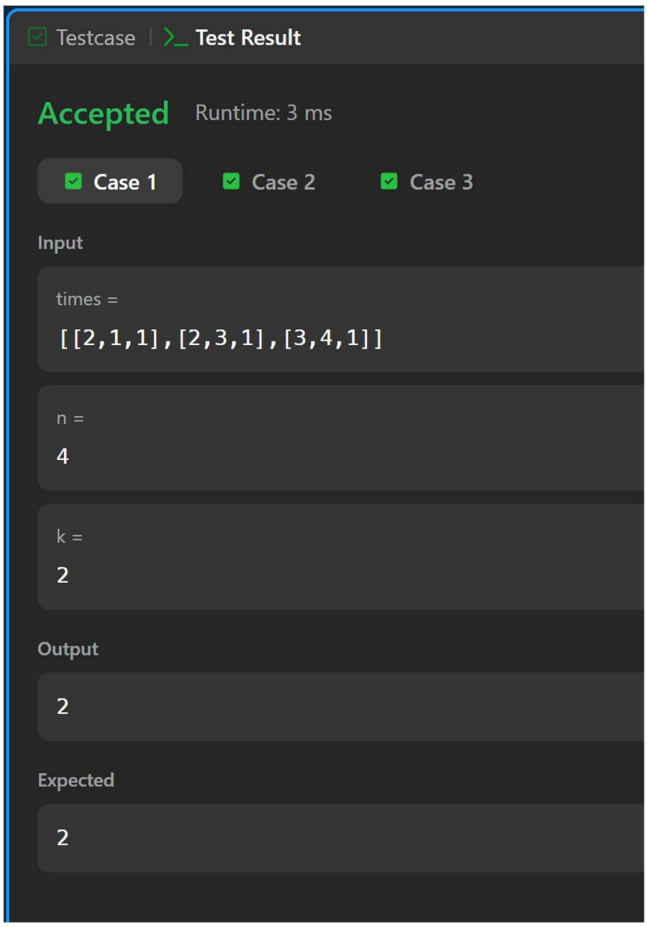
```
Enter number of vertices: 4
Enter adjacency matrix:
(Enter 99999 for INF)
0 5 99999 10
99999 0 3 99999
99999 99999 0 1
99999 99999 99999 0
Shortest Distance Matrix:
  0   5   8   9
INF   0   3   4
INF INF   0   1
INF INF INF   0

Process returned 0 (0x0)   execution time : 43.684 s
Press any key to continue.
```

LeetCode 743 - Network Delay Time

```
int networkDelayTime(int** times, int timesSize, int* timesColSize, int n, int k) {
    int dist[101];
    int visited[101] = {0};
    for(int i = 1; i <= n; i++)
        dist[i] = 1000000000;
    dist[k] = 0;
    for(int cnt = 0; cnt < n; cnt++) {
        int u = -1;
        for(int i = 1; i <= n; i++) {
            if(!visited[i] && (u == -1 || dist[i] < dist[u]))
                u = i;
        }
        visited[u] = 1;
        for(int i = 0; i < timesSize; i++) {
            int from = times[i][0];
            int to = times[i][1];
            int w = times[i][2];
            if(from == u && dist[u] + w < dist[to])
                dist[to] = dist[u] + w;
            }
        }
    }
    int ans = 0;
```

```
    for(int i = 1; i <= n; i++) {
        if(dist[i] == 1000000000)
            return -1;
        if(dist[i] > ans)
            ans = dist[i];
    }
    return ans;
}
```

OUTPUT:



Lab program 8:

a) **Find Minimum Cost Spanning Tree of a given undirected graph using Prim's algorithm.**

CODE:

```
#include <stdio.h>
#define MAX 10
#define INF 99999

int minKey(int key[], int mstSet[], int V) {
   int min = INF, minIndex;

   for(int i = 0; i < V; i++) {
      if(mstSet[i] == 0 && key[i] < min) {
         min = key[i];
         minIndex = i;
      }
   }
   return minIndex;
}

void primMST(int graph[MAX][MAX], int V) {
   int parent[MAX];
   int key[MAX];
   int mstSet[MAX];

   for(int i = 0; i < V; i++) {
      key[i] = INF;
      mstSet[i] = 0;
   }
   key[0] = 0;
   parent[0] = -1;

   for(int count = 0; count < V-1; count++) {
      int u = minKey(key, mstSet, V);
      mstSet[u] = 1;

      for(int v = 0; v < V; v++) {
         if(graph[u][v] && mstSet[v] == 0 && [u][v] < key[v]) {
            parent[v] = u;
            key[v] = graph[u][v];
         }
      }
   }
```

```
    int cost = 0;
    printf("\nEdges in MST:\n");
    printf("Edge\tWeight\n");

    for(int i = 1; i < V; i++) {
        printf("%d - %d\t%d\n", parent[i], i, graph[i][parent[i]]);
        cost += graph[i][parent[i]];
    }
    printf("\nMinimum Cost = %d\n", cost);
}

int main() {
    int V;
    int graph[MAX][MAX];
    printf("Enter number of vertices: ");
    scanf("%d", &V);
    printf("Enter adjacency matrix:\n");

    for(int i = 0; i < V; i++) {
        for(int j = 0; j < V; j++)
            scanf("%d", &graph[i][j]);
    }
    primMST(graph, V);
    return 0;
}
```

OUTPUT:

```
Enter number of vertices: 5
Enter adjacency matrix:
0 2 0 6 0
2 0 3 8 5
0 3 0 0 7
6 8 0 0 9
0 5 7 9 0

Edges in MST:
Edge    Weight
0 - 1   2
1 - 2   3
0 - 3   6
1 - 4   5

Minimum Cost = 16

Process returned 0 (0x0)   execution time : 44.066 s
Press any key to continue.
```

Lab program 8:

b) Find Minimum Cost Spanning Tree of a given undirected graph using Kruskal's algorithm.

CODE:

```
#include <stdio.h>
#define MAX 20

struct Edge {
   int u, v, w;
};

struct Edge edges[MAX];
int parent[MAX];

int find(int i) {
   while(parent[i] != i)
      i = parent[i];
   return i;
}

void unionSet(int a, int b) {
   parent[a] = b;
}

int main() {
   int V, E;

   printf("Enter number of vertices: ");
   scanf("%d", &V);

   printf("Enter number of edges: ");
   scanf("%d", &E);

   printf("Enter source destination weight:\n");

   for(int i = 0; i < E; i++)
      scanf("%d %d %d", edges[i].u, &edges[i].v, edges[i].w);

   for(int i = 0; i < V; i++)
      parent[i] = i;
```

```
    for(int i = 0; i < E - 1; i++) {
        for(int j = 0; j < E - i - 1; j++) {
            if(edges[j].w > edges[j + 1].w) {
                struct Edge temp = edges[j];
                edges[j] = edges[j + 1];
                edges[j + 1] = temp;
            }
        }
    }

    int count = 0;
    int cost = 0;
    printf("\nEdges in MST:\n");

    for(int i = 0; i < E && count < V - 1; i++) {
        int a = find(edges[i].u);
        int b = find(edges[i].v);

        if(a != b) {
            printf("%d -- %d = %d\n", edges[i].u, edges[i].v, edges[i].w);
            cost += edges[i].w;
            unionSet(a, b);
            count++;
        }
    }
    printf("\nMinimum Cost = %d\n", cost);
    return 0;
}
```

OUTPUT:

```
Enter number of vertices: 4
Enter number of edges: 5
Enter source destination weight:
0 1 10
0 2 6
0 3 5
1 3 15
2 3 4

Edges in MST:
2 -- 3 = 4
0 -- 3 = 5
0 -- 1 = 10

Minimum Cost = 19

Process returned 0 (0x0)   execution time : 38.116 s
Press any key to continue.
```

Lab program 9:
Implement Fractional Knapsack using Greedy technique.

CODE:

```
#include <stdio.h>

struct Item {
    int profit;
    int weight;
    float ratio;
};

int main() {
    int n, capacity;
    printf("Enter number of items: ");
    scanf("%d", &n);
    struct Item item[n];
    printf("Enter profit and weight of each item:\n");

    for(int i = 0; i < n; i++) {
        scanf("%d %d", &item[i].profit, &item[i].weight);
        item[i].ratio = (float)item[i].profit / item[i].weight;
    }

    printf("Enter knapsack capacity: ");
    scanf("%d", &capacity);

    for(int i = 0; i < n - 1; i++) {
        for(int j = 0; j < n - i - 1; j++) {
            if(item[j].ratio < item[j + 1].ratio) {
                struct Item temp = item[j];
                item[j] = item[j + 1];
                item[j + 1] = temp;
            }
        }
    }
    float totalProfit = 0.0;
```

```
    for(int i = 0; i < n; i++) {
        if(capacity >= item[i].weight) {
            totalProfit += item[i].profit;
            capacity -= item[i].weight;
        }
        else {
            totalProfit += item[i].ratio * capacity;
            break;
        }
    }

    printf("Maximum Profit = %.2f\n", totalProfit);

    return 0;
}
```

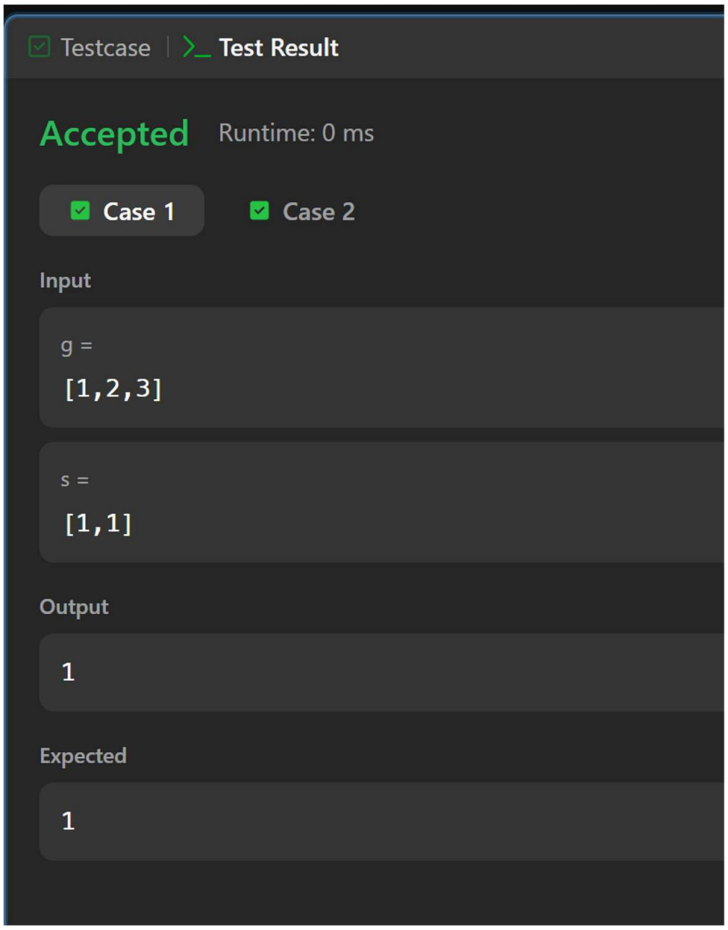
OUTPUT:

```
Enter number of items: 3
Enter profit and weight of each item:
60 10
100 20
120 30
Enter knapsack capacity: 50
Maximum Profit = 240.00

Process returned 0 (0x0)   execution time : 14.857 s
Press any key to continue.
```

LeetCode 455 - Assign Cookies

```
int findContentChildren(int* g, int gSize, int* s, int sSize) {
    for(int i=0;i<gSize-1;i++) {
        for(int j=0;j<gSize-i-1;j++) {
            if(g[j]>g[j+1]){
                int t=g[j];
                g[j]=g[j+1];
                g[j+1]=t;
            }
        }
    }
    for(int i=0;i<sSize-1;i++) {
        for(int j=0;j<sSize-i-1;j++) {
            if(s[j]>s[j+1]) {
                int t=s[j];
                s[j]=s[j+1];
                s[j+1]=t;
            }
        }
    }
    int i=0,j=0;
    while(i<gSize && j<sSize) {
        if(s[j]>=g[i])
            i++;
        j++;
    }
    return i;
}
```

OUTPUT:


Testcase
Test Result
Accepted
Runtime: 0 ms
Case 1
Case 2
Input
g =
[1,2,3]
s =
[1,1]
Output
1
Expected
1

Lab program 10:
From a given vertex in a weighted connected graph, find shortest paths to other vertices using Dijkstra's algorithm.

CODE:

```
#include <stdio.h>
#define MAX 10
#define INF 99999

int minDistance(int dist[], int visited[], int V) {
   int min = INF, minIndex;

   for(int i = 0; i < V; i++) {
      if(!visited[i] && dist[i] < min) {
         min = dist[i];
         minIndex = i;
      }
   }
   return minIndex;
}

void dijkstra(int graph[MAX][MAX], int V, int src) {
   int dist[MAX];
   int visited[MAX] = {0};

   for(int i = 0; i < V; i++)
      dist[i] = INF;

   dist[src] = 0;

   for(int count = 0; count < V - 1; count++) {
      int u = minDistance(dist, visited, V);
      visited[u] = 1;

      for(int v = 0; v < V; v++) {
         if(!visited[v] && graph[u][v] && dist[u] != INF && dist[u] + graph[u][v] < dist[v])
            dist[v] = dist[u] + graph[u][v];
      }
   }
   printf("\nVertex\tDistance from Source\n");

   for(int i = 0; i < V; i++)
      printf("%d\t%d\n", i, dist[i]);
}
```

```
int main() {
    int V, src;
    int graph[MAX][MAX];

    printf("Enter number of vertices: ");
    scanf("%d", &V);

    printf("Enter adjacency matrix:\n");

    for(int i = 0; i < V; i++) {
        for(int j = 0; j < V; j++)
            scanf("%d", &graph[i][j]);
    }
    printf("Enter source vertex: ");
    scanf("%d", &src);
    dijkstra(graph, V, src);

    return 0;
}
```

OUTPUT:

```
Enter number of vertices: 5
Enter adjacency matrix:
0 10 0 30 100
10 0 50 0 0
0 50 0 20 10
30 0 20 0 60
100 0 10 60 0
Enter source vertex: 0

Vertex  Distance from Source
0       0
1       10
2       50
3       30
4       60

Process returned 0 (0x0)   execution time : 39.772 s
Press any key to continue.
```

Lab program 11:
Implement "N-Queens Problem" using Backtracking.

CODE:

```
#include <stdio.h>
#define MAX 20
int board[MAX][MAX];

int isSafe(int row, int col, int n) {
   int i, j;

   for(i = 0; i < col; i++)
      if(board[row][i])
         return 0;

   for(i = row, j = col; i >= 0 && j >= 0; i--, j--)
      if(board[i][j])
         return 0;

   for(i = row, j = col; i < n && j >= 0; i++, j--)
      if(board[i][j])
         return 0;

   return 1;
}

int solveNQueens(int col, int n) {
   if(col >= n)
      return 1;

   for(int i = 0; i < n; i++) {
      if(isSafe(i, col, n)) {
         board[i][col] = 1;

         if(solveNQueens(col + 1, n))
            return 1;

         board[i][col] = 0;
      }
   }
   return 0;
}
```

```
int main() {
   int n;
   printf("Enter value of N: ");
   scanf("%d", &n);

   for(int i = 0; i < n; i++)
      for(int j = 0; j < n; j++)
         board[i][j] = 0;

   if(solveNQueens(0, n)) {
      printf("\nSolution:\n");

      for(int i = 0; i < n; i++) {
         for(int j = 0; j < n; j++)
            printf("%d ", board[i][j]);

         printf("\n");
      }
   }
   else
      printf("No solution exists\n");

   return 0;
}
```

OUTPUT: